

CT texture phantom dataset with paired image quality assessments for quantitative imaging

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Abstract

To establish a public repository of computed tomography (CT) texture phantom images paired with objective 3D image quality measurements from multiple scanner models using a diverse range of imaging protocols, facilitating investigations into relationships between image quality and quantitative imaging features. Three specialized CT phantoms were scanned: (1) the Corgi[®] phantom for image quality assessment, (2) a radiomics liver phantom, and (3) an open-source 3D-printed texture phantom. Image quality assessment included measurement of contrast-to-noise ratio, 3D modulation transfer function, and 3D noise power spectrum. Data were acquired on four CT scanner models from two manufacturers at five CTDI_{vol} levels (2.05–17.11 mGy) and reconstructed with eight different kernels, yielding 160 total conditions (combinations of scanner, dose, kernel). Data were validated for integrity and completeness, resulting in the exclusion of six conditions. The paired texture phantom and image quality (PTP-IQ) dataset includes: (1) DICOM image series of two texture phantoms acquired across the 154 conditions, as well as (2) image quality metrics derived from each corresponding set of scanner, acquisition and reconstruction settings provided in HDF5 format. This dataset enables the development and validation of harmonization methods for multi-center quantitative imaging studies, investigation of protocol-dependent QIF variability, and optimization of acquisition protocols for radiomics applications. The controlled and systematic study design facilitates isolation of individual protocol effects on quantitative measurements.

KEYWORDS

3D printing, CT image quality, CT phantoms, image quality, quantitative imaging, radiomics

1 | INTRODUCTION

Quantitative imaging features (QIFs) extracted from computed tomography (CT) images hold significant promise for precision medicine applications, including disease characterization, treatment response assessment, and outcome prediction.^{1–3} However, the clinical translation of QIF-based biomarkers faces substantial challenges due to variability introduced by differences in image acquisition and reconstruction protocols across scanners and institutions.^{4,5} This protocol-dependent variability can obscure true biological signals and compromise the reproducibility of radiomics studies.⁶

Texture-based QIFs, or radiomic texture features, have shown utility in predicting patient outcomes and treatment response.^{7–9} The relationship between CT image quality and image textures remains incompletely characterized, limiting efforts to develop effective harmonization strategies. While previous studies have demonstrated that acquisition parameters significantly influence texture features,^{10–13} systematic datasets enabling comprehensive investigation of these relationships are lacking. Existing phantom studies typically focus on either image quality assessment or texture analysis in isolation, preventing direct correlation between image quality and QIF variability.

To address this gap, we present a public dataset consisting of DICOM images of two texture phantoms with paired image quality measurements acquired under matched specified conditions across multiple CT scanner models and protocols. We include 3D measurements of the modulation transfer function (MTF) and the noise power spectrum (NPS), as well as contrast-to-noise ratio (CNR). Comprehensive quantification of 3D MTF and 3D NPS is essential for accurately characterizing the three-dimensional signal and noise transfer properties of modern CT systems.^{14,15} This dataset enables researchers to directly correlate these detailed image quality measurements with QIF measurements, particularly texture features that are inherently sensitive to image quality variations. This dataset is restricted to filtered back projection (FBP) reconstruction kernels. While iterative reconstruction is increasingly common in clinical practice, traditional image quality measurements may not fully characterize system performance under non-linear processing conditions, as measurements often become dependent on object contrast and noise levels.¹⁶

The primary objectives of this dataset are to: (1) provide a standardized resource for investigating protocol-dependent variability of image textures, (2) enable the development and validation of harmonization methods that account for underlying image quality differences, and (3) facilitate optimization of acquisition protocols for quantitative imaging applications. By making the paired texture phantom and image quality (PTP-IQ) dataset publicly available, we aim to accelerate progress toward robust, reproducible quantitative CT imaging.

2 | ACQUISITION AND VALIDATION METHODS

An image quality assessment phantom and two CT texture phantoms were scanned using matched conditions for 154 protocols across four scanners. The protocols capture a wide range of radiation dose levels and reconstruction kernels. Measurements of 3D spatial resolution, 3D noise characteristics, and CNR were made for each protocol.

This dataset consists of these image quality measurements and corresponding DICOM image series of the two texture phantoms for the 154 protocols.

2.1 | Phantoms

2.1.1 | Image quality phantom

Image quality characterization was performed using the Corgi[®] phantom (Phantom Laboratory, Salem, NY) that enables simultaneous measurement of CNR, 3D MTF, and 3D NPS using automated analysis soft-

ware (described in Section 2.3).¹⁷ Corgi[®] contains 13.5 cm, 4.0 cm thick cylindrical modules composed of polyurethane (PUR). For this study, a circular annulus was used making the outer phantom diameter 20.0 cm. Corgi[®] is comprised of blank modules which are uniform in composition, and test modules that are constructed with 20 mm diameter cylindrical inserts with hemispherical end-caps with known attenuation properties: polyvinyl chloride (PVC, 958 HU), polyoxymethylene (POM, 342 HU), polycarbonate (PC, 101 HU), nylon (PA, 31 HU), and polystyrene (PS, -27 HU) at 140 kV.¹⁸ The blank modules are used for 3D NPS measurements, and the test modules are used for 3D MTF and CNR measurements. Example images are shown in Figure 1.

The 3D measurement capability is particularly important for CT systems, as it accounts for potential anisotropy in spatial resolution and noise characteristics that may result from the helical acquisition geometry, variations in the effective focal spot size, or application of apodization kernel during reconstruction.^{14,15,19} This comprehensive approach provides a more complete characterization of image quality than traditional 2D measurements.

2.1.2 | Radiomics liver phantom

A radiomics liver phantom was used that was designed specifically for CT radiomics analysis (Phantom Laboratory, Salem, NY).²⁰ The phantom consists of a 20 cm diameter cylindrical body with three module layers comprised of a variety of patterned disk objects. The objects include solid and textured 3D printed patterns, and patterns composed of interspersed acetal shavings, 3 mm diameter polypropylene spheres, and fiberglass insulation with densities ranging from -30 to 120 HU. The 3D-printed objects were printed with acrylic then cast in clear urethane, providing a low-contrast texture object. Example images are shown in Figure 2. A detailed description and depictions of the phantom are provided in Varghese et al.²⁰

2.1.3 | Open-source 3D-printed phantom

An open-source 3D-printed phantom was used that was designed for quantitative CT imaging.²¹ The design files and assembly instructions are available at <https://zenodo.org/doi/10.5281/zenodo.15733372>.²² The phantom consists of a cylindrical shell (20 cm diameter × 10 cm height) printed with polylactic acid (PLA) filament and filled with epoxy resin. A PLA body ring was used to increase the total diameter of the phantom to 27.5 cm.

The phantom contains 3 cm diameter cylindrical modular inserts that can be customized to create experiment-specific test objects. The eight inserts used

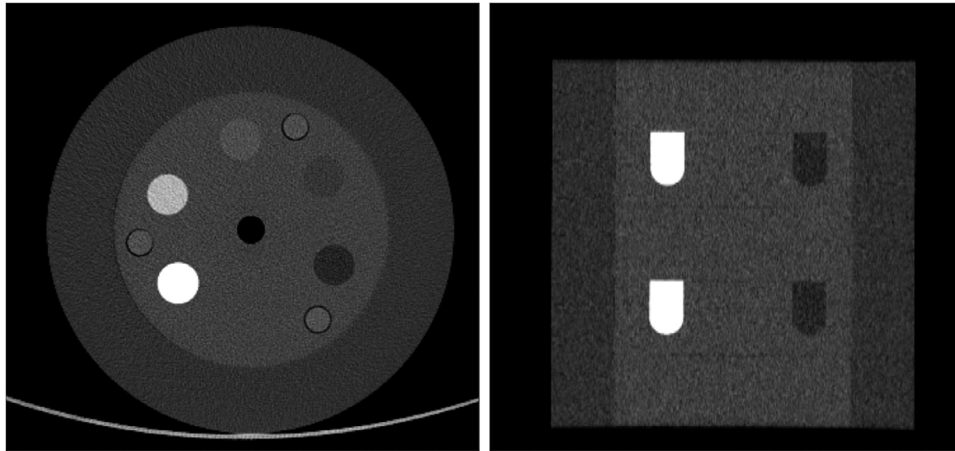


FIGURE 1 Example axial and coronal slices of the Corgi® phantom, used to measure image quality. Shown with window width 650 HU, window level 175 HU.



FIGURE 2 Example slices of the radiomics liver phantom. Shown with window width 250 HU, window level 50 HU.

were all uniform in the z-direction and composed of combinations of two densities (3D printed PLA, and air as the background material). The phantom with the texture objects labeled 1-8 is shown in Figure 3. Inserts 1, 2, and 3 were uniformly spaced 3 mm circular rods with varied pitch (4, 6, 8 mm). Inserts 4 and 5 were checkerboard patterns with different scales: grid square side lengths of 3 mm and 7 mm. Inserts 6 and 7 were spiculated solid objects of different scales. Insert 8 was a texture composed of circles with random size and positioning. The positioning and orientation of the texture inserts within the phantom was kept as shown for all acquisitions.

This phantom's relatively simple geometric texture patterns are not intended to mimic specific biological tissues or clinical pathologies.

2.2 | Image acquisition

Image acquisition was performed at two academic medical centers using four CT scanner models from two manufacturers: SOMATOM Force and SOMATOM

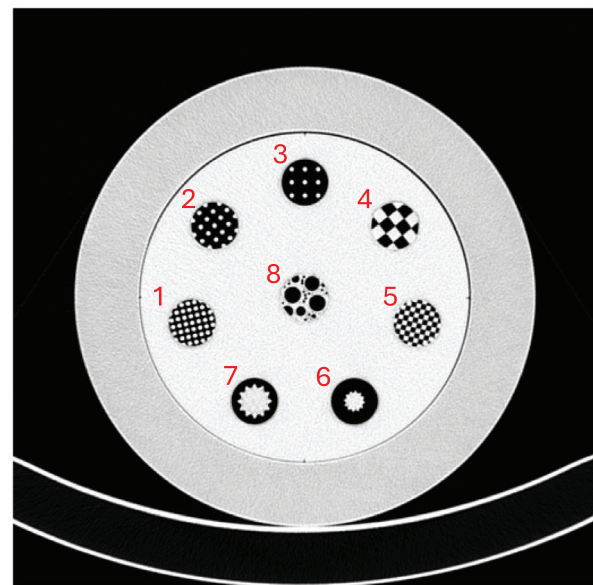


FIGURE 3 The 3D-printed texture phantom with the texture insert objects labeled 1-8. Shown with window width 1300 HU, window level -450 HU.

TABLE 1 Image acquisition and reconstruction conditions for each scanner.

Scanner model	CTDI _{vol} (mGy)	Convolution kernel
Revolution Apex	2.05, 4.29, 6.53, 10.65, 17.05	BONE, BONEPLUS, CHST, DETAIL, EDGE, LUNG, SOFT, STANDARD
Revolution EVO	2.21, 4.20, 6.19, 10.62, 17.03	BONE, BONEPLUS, CHST, DETAIL, EDGE, LUNG, STANDARD
Definition AS	2.05, 4.10, 6.38, 10.49, 16.87	Br32f, Br36f, Br40f, Br43f, Br49f, Br54f, Br59f, Br62f
Force	2.14, 4.28, 6.42, 10.70, 17.11	Br32d, Br36d, Br40d, Br44d, Br49d, Br54d, Br59d, Br64d

Note: The Force, 4.28 mGy, Br54d condition was not included in the final dataset; all other combinations of CTDI_{vol} and kernel listed were included. The comprehensive list of protocol details can be found in Appendix A.

Definition AS64 (Siemens Healthineers, Forchheim, Germany) and Revolution EVO and Revolution Apex (GE Healthcare, Waukesha, Wisconsin).

The image quality phantom and the two texture phantoms were scanned under matched conditions with careful alignment for a diverse set of CT protocols. Scans were acquired at five CTDI_{vol} levels and reconstructed with eight reconstruction kernels. This was repeated across the four different CT scanner systems. All other parameters were kept constant across acquisitions from the same scanner, and matched as closely as possible between scanner models. Imaging conditions were selected with the goal of capturing varied image quality (i.e., noise and spatial resolution) within a range of clinical practice for chest and abdomen and pelvis CT examinations. We did not specifically consider head and neck CT examinations in this protocol selection.

The CTDI_{vol} levels ranged from 2.1 mGy to 17.1 mGy. The CTDI_{vol} values were selected to approximate that of the lowest dose CT examinations (lung cancer screening) through that of more typical diagnostic CT examinations (chest or abdomen and pelvis, depending on patient size).^{23,24} This range will capture a range of clinically realizable noise conditions. All scans used 120 kV; the desired CTDI_{vol} levels were obtained by varying only tube current.

Reconstruction kernels were chosen to capture a range of clinically relevant FBP reconstructions. For the GE scanners, this included all eight kernels available. For the Siemens scanners, eight kernels were selected that ranged from a smoothing kernel to a sharp over-enhancing kernel. All reconstructions employed FBP algorithms. The kernels and dose levels selected for each scanner model can be found in Table 1.

Scans were acquired with a slice thickness of 1.0 mm on Siemens scanners and 1.25 mm on GE scanners, as these were the closest thin slice thicknesses available. Thin slice thickness was selected as a higher resolution is generally preferred for quantitative imaging to reduce partial volume effects. Additionally, the provision of thin-slice data offers maximum flexibility, as thick-slice datasets (e.g., 2.5 mm or 5.0 mm) can be readily synthesized from thin-slice images through simple averaging. This allows users to simulate various clinical viewing conditions from the provided data. Conversely, generat-

ing thin slices from thick-slice acquisitions is not possible without the loss of high-frequency information or the introduction of interpolation artifacts.

All protocols used 120 kV, 0.5 second rotation time, pitch of 1.0, and 350 mm field of view. Tube current modulation was disabled to ensure that acquisition was consistent across slices, and enable precise and reproducible pairing between texture phantom scans and image quality measurements. Detailed descriptions of the four CT scanner systems and the parameters that were held constant across acquisitions are provided in Table 2.

A small subset of DICOM data, specifically the Corgi phantom with the EVO SOFT kernel and the texture phantoms with the Force Br54d kernel at 4.28 mGy, was excluded from the final dataset due to unrecoverable file corruption and data transfer errors. Because we were not able to recover the original files, these specific conditions were excluded and the final dataset consisted of 154 CT protocols. The details of each protocol included in the dataset can be found in Appendix A.

2.3 | Image quality measurements

Image quality was measured using the Corgi[®] phantom. The 3D MTF, 3D NPS, and CNR were computed using automated software evaluation as described in Siewerden and Uneri et al. (2020).¹⁷

Spatial resolution was measured both in-plane (also referred to as “axial” or “2D”) and out-of-plane (“oblique” or “3D”) using the PVC insert. The edge spread function (ESF) was measured and used to calculate the MTF for the 2D and 3D measurements. The MTF measurements were made on the highest dose scans to minimize uncertainty introduced by image noise. For a given scanner system and kernel, the spatial resolution measurements made on the highest dose scan were then propagated to the lower dose scans. For linear FBP, it is reasonable to assume that MTF is independent of dose.

The 3D NPS was measured on the phantom’s uniform module, composed of homogeneous PUR. The NPS curves are obtained by radially sampling the 3D NPS in the axial and oblique directions. The NPS was measured for each of the 154 conditions.

TABLE 2 The CT scanner details and image acquisition parameters used for each of the four systems used. These parameters were used consistently for all acquisitions for all phantoms.

Parameter	DICOM tag	Scanner 1	Scanner 2	Scanner 3	Scanner 4
Manufacturer	(0008, 0070)	SIEMENS	SIEMENS	GE MEDICAL SYSTEMS	GE MEDICAL SYSTEMS
Model	(0008, 1090)	SOMATOM Definition AS	SOMATOM Force	Revolution EVO	Revolution Apex
Software Version	(0018, 1020)	syngo CT VB20A	syngo CT VB20A	cj_kl.89	cadence_ct_25.44
Slice thickness (mm)	(0018, 0050)	1.0	1.0	1.25	1.25
Tube voltage (kV)	(0018, 0060)	120	120	120	120
Reconstruction diameter (mm)	(0018, 1100)	350	350	350	350
Rows	(0028, 0010)	512	512	512	512
Columns	(0028, 0011)	512	512	512	512
Pixel Spacing	(0028, 0030)	0.68	0.68	0.68	0.68

The CNR measurements were made on each insert (PVC, POM, PC, PA, PS) to assess contrast detectability across a range of attenuations. The measured attenuation, contrast, and CNR are reported for each insert. The CNR was measured for each of the 154 conditions.

2.4 | Data validation

All data components were processed through an automated pipeline utilizing metadata from DICOM headers for image series and HDF5 attributes for image quality metrics. Data completeness was validated by ensuring a 1:1 mapping between each texture phantom DICOM series and its corresponding image quality dataset for all 154 included protocols. A protocol was only finalized if complete data existed for both texture phantoms and the associated image quality measurements.

The DICOM image data from the two texture phantoms were manually reviewed for artifacts, alignment, and accurate protocol labeling. Figure 4 shows example images of the two texture phantoms from each of the four scanner systems.

Image quality measurements were compared across protocols and plotted to verify data integrity, as well as to verify that measurements were generally consistent with expected behavior. For example, we observed that the noise magnitude increased with decreasing dose, and MTF curves shifted to higher frequencies with sharper reconstruction kernels. Relative trends with dose and kernel selection were consistent, validating the protocol design.

The measured image quality curves from the highest dose protocols of the dataset ($\text{CTDI}_{vol} \approx 17$) are shown in Figure 5. Evaluating a single dose level is sufficient to capture the majority of the dataset's range in image quality, as varying dose does not impact MTF

or noise texture (only noise magnitude). The measured axial and oblique MTF curves demonstrate a wide range of spatial resolution, consistent with the range of reconstruction kernels included in the dataset (ranging from smoothing to sharpening). Peak MTF values exceeding 1.0 indicate edge-enhancing reconstruction. The oblique MTF curves (middle) are compressed to lower spatial frequencies compared to the axial MTF curves (top), reflecting the reduced resolution in the slice-thickness direction relative to the in-plane direction. The NPS curves (bottom) illustrate the expected tradeoff between spatial resolution and noise: sharper kernels, which yield higher spatial resolution, are associated with substantially elevated noise magnitude and a shift of noise power toward higher spatial frequencies. Line style denotes scanner model (see legend), with multiple curves per scanner corresponding to the set of reconstruction kernels measured; the four scanners span a broadly comparable range of imaging conditions. Taken together, these curves confirm that the imaging conditions sampled span a meaningful range of the resolution-noise tradeoff space, providing a diverse set of protocols for evaluating relationships between image quality and texture images.

3 | DATA FORMAT AND USAGE NOTES

The dataset consists of DICOM image series for two texture phantoms, HDF5 files containing image quality data for 154 CT protocols, and a summary metadata file. All files have been made available via Zenodo (<https://doi.org/10.5281/zenodo.16820900>). The total dataset size is 17.57 GB uncompressed. Prior to upload, each sub-directory was compressed using gzip (<https://www.gnu.org/software/gzip/>), after which the total size was 9.59 GB. The repository is organized as follows:

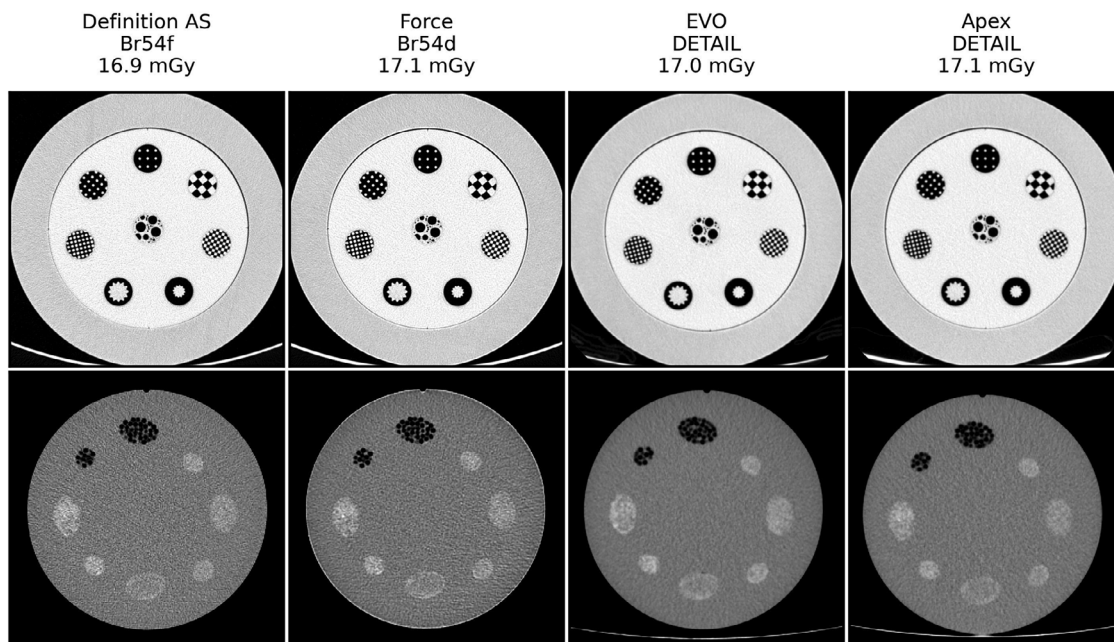


FIGURE 4 Example CT images of the two texture phantoms. Example CT images of the two texture phantoms (top row: 3D printed phantom, bottom row: radiomics liver phantom) from each of the four scanner models, with protocol labeled above. Images cropped for visualization. Top row shown with WW: 1100 HU, WL: -400 HU; bottom row shown with WW: 250 HU, WL: 50 HU.

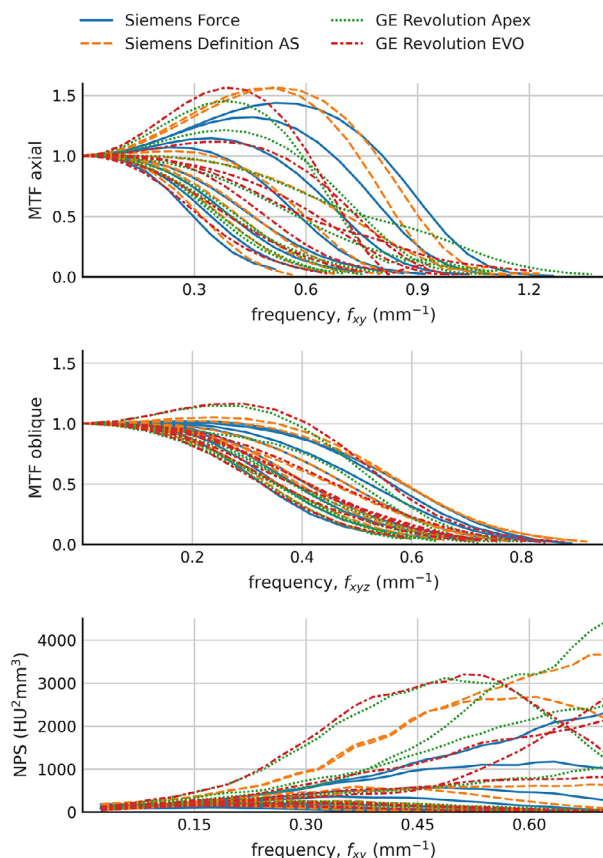


FIGURE 5 Measured image quality data (MTF and NPS). Measured image quality data from the highest CTDI_{vol} protocols of the dataset (≈ 17 mGy). Each line corresponds to a single imaging condition. From top to bottom: axial MTF, oblique MTF, axial NPS.

```
ptp-iq-dataset/
├── metadata.csv
├── radiomics-liver-phantom/
│   ├── {protocol_id}/
│   │   └── {instance_number}.dcm
├── 3d-printed-texture-phantom/
│   ├── {protocol_id}/
│   │   └── {instance_number}.dcm
├── image-quality-data/
│   └── {protocol_id}.h5
├── examples/
│   ├── example-data/
│   ├── environment.yaml
│   └── data_reading_example.ipynb
```

The use of the DICOM data format, HDF5 file format, gzip compression, and Zenodo repository align with Findable, Accessible, Interoperable, Reusable (FAIR) data management principles.

3.1 | Image data

All texture phantom image data are provided in DICOM format, compliant with the DICOM standard. Files retain the original header information including complete acquisition and reconstruction parameters.

Free and open source toolkits are available to support the programmatic access of DICOM data. Software

TABLE 3 Description of the contents of the HDF5 files.

Key in *.h5 file	Description	Units
app/version	Image quality extraction software version	—
image/dicom/{dicom_tag}	DICOM header metadata extracted from the corresponding Corgi® phantom image series	—
cnr1/air/intensity	Mean HU of air	HU
cnr1/pur/intensity	Mean HU of acrylic background	HU
cnr1/{pvc pom pc palps}/intensity	Mean HU of {material}	HU
cnr1/{pvc pom pc palps}/contrast	Contrast of {material}	HU
cnr1/{pvc pom pc palps}/cnr	CNR of {material}	1
mtf1/pvc/{mtf2d mtf3d}/esf_x	{Axi. Obl.} ESF x values	mm
mtf1/pvc/{mtf2d mtf3d}/esf_y	{Axi. Obl.} ESF y values	HU
mtf1/pvc/{mtf2d mtf3d}/sesf	{Axi. Obl.} ESF σ	mm
mtf1/pvc/{mtf2d mtf3d}/mtf_x	{Axi. Obl.} MTF x values	mm ⁻¹
mtf1/pvc/{mtf2d mtf3d}/mtf_y	{Axi. Obl.} MTF y values	1
mtf1/pvc/{mtf2d mtf3d}/f10	Freq. MTF falls to 10%	mm ⁻¹
mtf1/pvc/{mtf2d mtf3d}/f50	Freq. MTF falls to 50%	mm ⁻¹
nps1/noise	Image noise (18 ROIs)	HU
nps1/nps	3D NPS	HU ² mm ³
nps1/{nps2d nps3d}/nps_x	{Axi. Obl.} NPS x values	mm ⁻¹
nps1/{nps2d nps3d}/nps_y	{Axi. Obl.} NPS y values	HU ² mm ³

packages that provide this functionality include DCMTK in C++ (<https://git.dcmctk.org/>), *pydicom* in Python (<https://github.com/pydicom/pydicom>), and *dicom* in Golang (<https://github.com/suyashkumar/dicom>). Radiomics analysis is provided by the Python package *pyradiomics* (<http://github.com/radiomics/pyradiomics>).

There are no specific recommendations for preprocessing steps, as this decision is highly dependent on the specific intended use case.

3.2 | Image quality data

The image quality data is provided as HDF5 files, which provides a way to store a variety of heterogeneous data objects. The data includes measurements made describing CNR, MTF, and NPS. The DICOM header metadata from the corresponding CT image of the Corgi® phantom is also included in the HDF5 files. Each of the 154 protocols has a corresponding HDF5 file with image quality measurements. The contents of the file is shown in Table 3.

Several packages are available for interfacing with the HDF5 binary data format. The HDF Group provides a visual tool for browsing and editing HDF5 files called *HDFView* (<https://www.hdfgroup.org/download-hdfview/>). An online HDF5 file viewing tool provided by the European Synchrotron Radiation Facility called *myHDF5* (<https://myhdf5.hdfgroup.org/>) includes some data visualization, such as line plots and heatmap views of arrays. An example of the visualization provided

by this tool is shown in Figure 6. The *h5py* package enables straightforward manipulation of the data in Python (<https://github.com/h5py/h5py>).

4 | DISCUSSION

This dataset provides a unique resource for the quantitative imaging community by enabling direct investigation of the relationship between fundamental image quality metrics and images textures. The systematic variation of acquisition parameters across multiple scanner platforms creates a controlled experimental framework for addressing key challenges in reproducibility and robustness for quantitative imaging.

One important application of this dataset is in feature robustness analysis, where image features (e.g., radiomics or handcrafted features) can be extracted and evaluated for stability across a range of acquisition and reconstruction protocols. This can inform the selection of QIFs that are reproducible for some relevant range of protocols. Conversely, the dataset may also inform protocol optimization for specific quantitative imaging tasks: if particular features are known to be predictive at a given operating point, one could determine the protocol conditions under which those features remain stable, guiding protocol recommendations tailored to a given clinical application.

More specifically, this dataset enables the analysis of feature robustness and reproducibility in terms of measurable descriptors of image quality, rather than

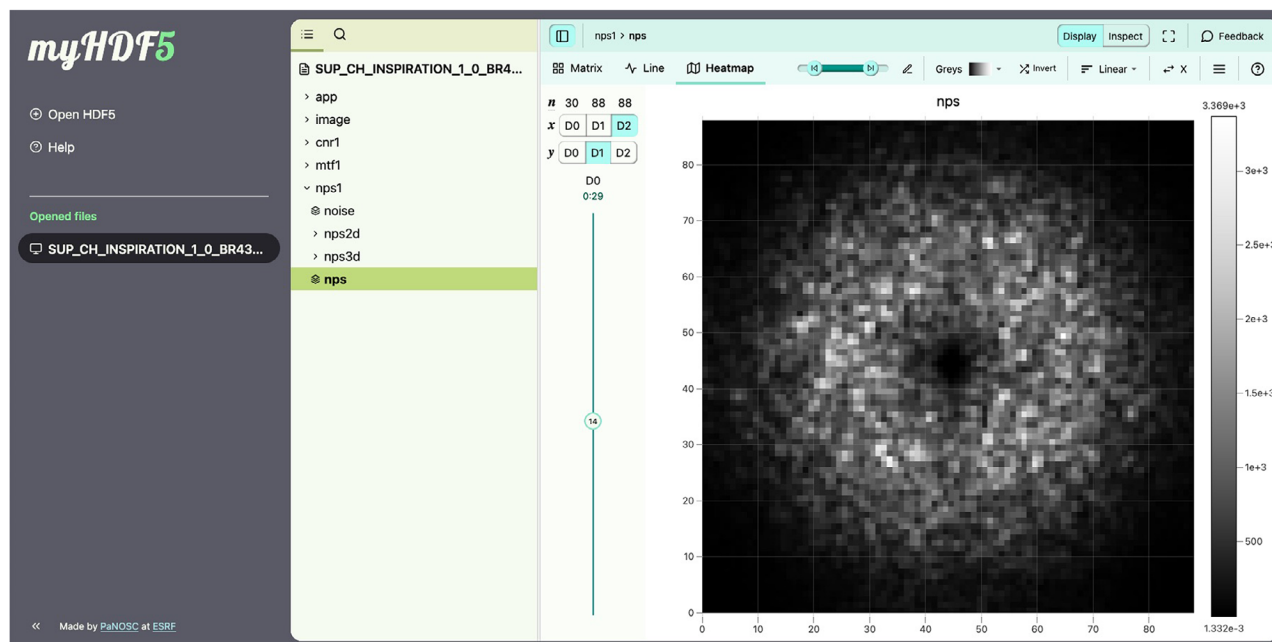


FIGURE 6 Example visualization of the image quality data using the *myHDF5* web viewer tool. Shown is a slice of the 3D NPS for the Siemens SOMATOM Definition AS64, 4.1 mGy, Br43f protocol.

relying solely on protocol descriptions that depend on manufacturer-specific terminology. While image quality measurements themselves are the multi-factorial result of radiation dose, reconstruction kernel, and vendor-specific system design, they provide a common, quantitative basis for characterizing the imaging conditions.

Currently, the reliance on categorical labels (e.g., ‘Br40d’ vs. ‘Bone’ kernels) limits the generalizability of radiomic studies, as these labels lack a physical definition and vary significantly between manufacturers. We hypothesize that physical image quality measurements provide a standardized, manufacturer-independent coordinate system for comparing image acquisition and reconstruction conditions. By studying radiomic features within this physical space, we collapse the high-dimensional variability of proprietary hardware and software into a continuous set of comparable metrics that will enable a more direct translation and application of research results between studies. These approaches could be particularly important for achieving multi-center clinical deployment of quantitative imaging-based predictive models; an objective that has remained challenging despite promising results in research studies.

In practice, the image quality metrics provided in this dataset could be incorporated into harmonization pipelines in several ways. Physical metrics such as MTF and NPS have been proposed as a basis for harmonization through prospective protocol selection as a way of reducing variation in radiomic features across scanner models.^{25,26} These metrics have also been proposed as a basis for analytically predicting

and correcting protocol-induced variation in radiomic features. In this framework, MTF and NPS serve as quantitative inputs that characterize the resolution and noise state of each imaging condition, enabling feature-level corrections that are grounded in the underlying physics rather than dependent on categorical protocol labels.^{27,28} The DICOM image subset alone also constitutes a valuable resource for validating or benchmarking image-based harmonization and domain adaptation techniques: because the physical object being scanned is held constant across all imaging conditions, protocol-induced variation is the sole source of differences between images, providing a degree of experimental control not achievable with clinical data.^{29–31}

A number of potential extensions to this dataset are possible. Future contributions and collaborations are supported by hosting on Zenodo, a stable open-access repository maintained by CERN, which provides persistent archival and a permanent DOI to ensure the long-term accessibility of this dataset. For example, external research groups could contribute annotations or region-of-interest segmentations for the texture phantoms. This could be extended to the release of the associated pre-computed radiomic feature measurements to supplement the existing data.

Extending the dataset to additional texture objects is possible if a group has access to one of the scanners used in the dataset. With careful matching of the protocol, the dataset’s image quality data could be directly applied to study different texture phantoms. As the phantoms used in this study consist of relatively simple texture patterns, extending to texture patterns representing specific pathologies, or anthropomorphic

phantom configurations for anatomical realism could be of interest for targeted clinical investigations. For example, PixelPrint enables the 3D printing of patient-specific lung phantoms with accurate attenuation profiles and textures.³²

This dataset differs from previously published CT phantom datasets in several important respects. For example, Kalendralis et al. released a multicenter CT phantom dataset intended for radiomics reproducibility tests.³³ However, Kalendralis et al. dataset used the COPDGene Phantom II (Phantom Laboratory, Salem, NY) and the Abdominal Phantom (CIRS, Norfolk, VA) which are composed of uniform density materials and were not designed for radiomics studies. Additionally, their dataset does not include image quality measurements. The Credence Cartridge Radiomics (CCR) Phantom dataset includes a variety of textures and was explicitly crafted to study radiomics features.⁴ However, it also lacks accompanying measurements of image quality metrics, limiting its utility for studies that seek to correlate texture-based features with objective image quality.

There are several limitations of the dataset that should be acknowledged. First, while the dataset captures protocol variability across scanner models, manufacturers, dose levels, and reconstruction kernels, other sources of variability were not systematically explored. These factors, such as slice thickness, beam energy, patient size, or positioning, can also significantly influence image appearance and feature values and should be considered in future studies.

The phantoms used in this study represent a limited set of texture patterns. While they provide a controlled and reproducible means of assessing texture sensitivity, they may not fully capture the complexity and heterogeneity found in biological tissues. There are currently no standard reference objects for assessing image texture in CT, and most commercially available phantoms are composed of uniform materials. The texture objects included in this dataset represent a preliminary step toward filling that gap, but additional efforts are needed to develop and standardize texture phantoms for quantitative imaging.

This dataset does not include images reconstructed using iterative reconstruction algorithms. Although iterative methods are widely used in clinical practice, their non-linear behavior makes it more challenging to characterize in terms of image quality. As connecting image data to corresponding quantitative image quality measurements was fundamental to the design of this dataset, and because iterative and deep learning reconstruction methods can alter image texture and noise characteristics in ways that are not captured by traditional image quality metrics, the dataset is most directly applicable to FBP reconstructions. Future extensions of the dataset may consider inclusion of iterative reconstructions, along with additional measures of image

quality required to characterize them, such as contrast- and noise-dependent measures of spatial resolution, noise nonuniformity index, and noise inhomogeneity index; this may require defining a task-specific context.¹⁶

5 | CONCLUSION

We have provided a public dataset of CT images from two texture phantoms that have been acquired from multiple scanner models and a wide range of acquisition protocols, along with the associated image quality measurements from each of those scanner models and protocols. This unique resource enables systematic investigation of relationships between fundamental image quality metrics and quantitative imaging features, addressing a critical need in radiomics harmonization research. The dataset's standardized acquisition design, paired image quality measurements, and open-source components facilitate reproducible research toward robust quantitative CT imaging. By making the PTP-IQ dataset freely available, we aim to accelerate development of characterization and harmonization methods essential for clinical translation of QIF-based biomarkers.

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CONFLICT OF INTEREST STATEMENT

Andrew Hernandez has received research funding from the NIH (5-U01-EB034205-02), and is a paid consultant and stock options holder for a company (Izotropic Corporation) that is commercializing breast CT. UC Davis has research agreements with Canon Medical Systems and United Imaging Healthcare. Ali Uneri has received research funding from the NIH (R01EB031958, R01EB035559) and Korea Institute for Advancement of Technology (KIAT), and has a research agreement with Medtronic. Joshua Levy is an employee and Founder of The Phantom Laboratory, Salem, NY. UCLA has a Master Research Agreement with Siemens Healthineers (Forchheim, Germany); Michael McNitt-Gray has received research funding unrelated to this project under this agreement. The remaining authors have no relevant conflict of interest to disclose.

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APPENDIX A

Tables A.1–A.4

TABLE A.1 Image acquisition and reconstruction protocols for the GE Revolution Apex.

ID	Institution	Dose label	CTDI _{vol} (mGy)	Conv. kernel	Slice thickness (mm)
A-01	UC Davis	1	2.05	BONE	1.25
A-02	UC Davis	1	2.05	BONEPLUS	1.25
A-03	UC Davis	1	2.05	CHST	1.25
A-04	UC Davis	1	2.05	DETAIL	1.25
A-05	UC Davis	1	2.05	EDGE	1.25
A-06	UC Davis	1	2.05	LUNG	1.25
A-07	UC Davis	1	2.05	SOFT	1.25
A-08	UC Davis	1	2.05	STANDARD	1.25
A-09	UC Davis	2	4.29	BONE	1.25
A-10	UC Davis	2	4.29	BONEPLUS	1.25
A-11	UC Davis	2	4.29	CHST	1.25
A-12	UC Davis	2	4.29	DETAIL	1.25
A-13	UC Davis	2	4.29	EDGE	1.25
A-14	UC Davis	2	4.29	LUNG	1.25
A-15	UC Davis	2	4.29	SOFT	1.25
A-16	UC Davis	2	4.29	STANDARD	1.25
A-17	UC Davis	3	6.53	BONE	1.25
A-18	UC Davis	3	6.53	BONEPLUS	1.25
A-19	UC Davis	3	6.53	CHST	1.25
A-20	UC Davis	3	6.53	DETAIL	1.25
A-21	UC Davis	3	6.53	EDGE	1.25
A-22	UC Davis	3	6.53	LUNG	1.25
A-23	UC Davis	3	6.53	SOFT	1.25
A-24	UC Davis	3	6.53	STANDARD	1.25
A-25	UC Davis	4	10.65	BONE	1.25
A-26	UC Davis	4	10.65	BONEPLUS	1.25
A-27	UC Davis	4	10.65	CHST	1.25
A-28	UC Davis	4	10.65	DETAIL	1.25
A-29	UC Davis	4	10.65	EDGE	1.25
A-30	UC Davis	4	10.65	LUNG	1.25
A-31	UC Davis	4	10.65	SOFT	1.25
A-32	UC Davis	4	10.65	STANDARD	1.25
A-33	UC Davis	5	17.05	BONE	1.25
A-34	UC Davis	5	17.05	BONEPLUS	1.25
A-35	UC Davis	5	17.05	CHST	1.25
A-36	UC Davis	5	17.05	DETAIL	1.25
A-37	UC Davis	5	17.05	EDGE	1.25
A-38	UC Davis	5	17.05	LUNG	1.25
A-39	UC Davis	5	17.05	SOFT	1.25
A-40	UC Davis	5	17.05	STANDARD	1.25

TABLE A.2 Image acquisition and reconstruction protocols for the GE Revolution EVO.

ID	Institution	Dose label	CTDI _{vol} (mGy)	Conv. kernel	Slice thickness (mm)
E-01	UC Davis	1	2.21	BONE	1.25
E-02	UC Davis	1	2.21	BONEPLUS	1.25
E-03	UC Davis	1	2.21	CHST	1.25
E-04	UC Davis	1	2.21	DETAIL	1.25
E-05	UC Davis	1	2.21	EDGE	1.25
E-06	UC Davis	1	2.21	LUNG	1.25
E-07	UC Davis	1	2.21	STANDARD	1.25
E-08	UC Davis	2	4.20	BONE	1.25
E-09	UC Davis	2	4.20	BONEPLUS	1.25
E-10	UC Davis	2	4.20	CHST	1.25
E-11	UC Davis	2	4.20	DETAIL	1.25
E-12	UC Davis	2	4.20	EDGE	1.25
E-13	UC Davis	2	4.20	LUNG	1.25
E-14	UC Davis	2	4.20	STANDARD	1.25
E-15	UC Davis	3	6.19	BONE	1.25
E-16	UC Davis	3	6.19	BONEPLUS	1.25
E-17	UC Davis	3	6.19	CHST	1.25
E-18	UC Davis	3	6.19	DETAIL	1.25
E-19	UC Davis	3	6.19	EDGE	1.25
E-20	UC Davis	3	6.19	LUNG	1.25
E-21	UC Davis	3	6.19	STANDARD	1.25
E-22	UC Davis	4	10.62	BONE	1.25
E-23	UC Davis	4	10.62	BONEPLUS	1.25
E-24	UC Davis	4	10.62	CHST	1.25
E-25	UC Davis	4	10.62	DETAIL	1.25
E-26	UC Davis	4	10.62	EDGE	1.25
E-27	UC Davis	4	10.62	LUNG	1.25
E-28	UC Davis	4	10.62	STANDARD	1.25
E-29	UC Davis	5	17.03	BONE	1.25
E-30	UC Davis	5	17.03	BONEPLUS	1.25
E-31	UC Davis	5	17.03	CHST	1.25
E-32	UC Davis	5	17.03	DETAIL	1.25
E-33	UC Davis	5	17.03	EDGE	1.25
E-34	UC Davis	5	17.03	LUNG	1.25
E-35	UC Davis	5	17.03	STANDARD	1.25

TABLE A.3 Image acquisition and reconstruction protocols for the Siemens SOMATOM Definition AS.

ID	Institution	Dose label	CTDI _{vol} (mGy)	Conv. kernel	Slice thickness (mm)
D-01	UCLA	1	2.05	Br32f	1.00
D-02	UCLA	1	2.05	Br36f	1.00
D-03	UCLA	1	2.05	Br40f	1.00
D-04	UCLA	1	2.05	Br43f	1.00
D-05	UCLA	1	2.05	Br49f	1.00
D-06	UCLA	1	2.05	Br54f	1.00
D-07	UCLA	1	2.05	Br59f	1.00
D-08	UCLA	1	2.05	Br62f	1.00
D-09	UCLA	2	4.10	Br32f	1.00
D-10	UCLA	2	4.10	Br36f	1.00
D-11	UCLA	2	4.10	Br40f	1.00
D-12	UCLA	2	4.10	Br43f	1.00
D-13	UCLA	2	4.10	Br49f	1.00
D-14	UCLA	2	4.10	Br54f	1.00
D-15	UCLA	2	4.10	Br59f	1.00
D-16	UCLA	2	4.10	Br62f	1.00
D-17	UCLA	3	6.38	Br32f	1.00
D-18	UCLA	3	6.38	Br36f	1.00
D-19	UCLA	3	6.38	Br40f	1.00
D-20	UCLA	3	6.38	Br43f	1.00
D-21	UCLA	3	6.38	Br49f	1.00
D-22	UCLA	3	6.38	Br54f	1.00
D-23	UCLA	3	6.38	Br59f	1.00
D-24	UCLA	3	6.38	Br62f	1.00
D-25	UCLA	4	10.49	Br32f	1.00
D-26	UCLA	4	10.49	Br36f	1.00
D-27	UCLA	4	10.49	Br40f	1.00
D-28	UCLA	4	10.49	Br43f	1.00
D-29	UCLA	4	10.49	Br49f	1.00
D-30	UCLA	4	10.49	Br54f	1.00
D-31	UCLA	4	10.49	Br59f	1.00
D-32	UCLA	4	10.49	Br62f	1.00
D-33	UCLA	5	16.87	Br32f	1.00
D-34	UCLA	5	16.87	Br36f	1.00
D-35	UCLA	5	16.87	Br40f	1.00
D-36	UCLA	5	16.87	Br43f	1.00
D-37	UCLA	5	16.87	Br49f	1.00
D-38	UCLA	5	16.87	Br54f	1.00
D-39	UCLA	5	16.87	Br59f	1.00
D-40	UCLA	5	16.87	Br62f	1.00

TABLE A.4 Image acquisition and reconstruction protocols for the Siemens SOMATOM Force.

ID	Institution	Dose label	CTDI _{vol} (mGy)	Conv. kernel	Slice thickness (mm)
F-01	UCLA	1	2.14	Br32d	1.00
F-02	UCLA	1	2.14	Br36d	1.00
F-03	UCLA	1	2.14	Br40d	1.00
F-04	UCLA	1	2.14	Br44d	1.00
F-05	UCLA	1	2.14	Br49d	1.00
F-06	UCLA	1	2.14	Br54d	1.00
F-07	UCLA	1	2.14	Br59d	1.00
F-08	UCLA	1	2.14	Br64d	1.00
F-09	UCLA	2	4.28	Br32d	1.00
F-10	UCLA	2	4.28	Br36d	1.00
F-11	UCLA	2	4.28	Br40d	1.00
F-12	UCLA	2	4.28	Br44d	1.00
F-13	UCLA	2	4.28	Br49d	1.00
F-14	UCLA	2	4.28	Br59d	1.00
F-15	UCLA	2	4.28	Br64d	1.00
F-16	UCLA	3	6.42	Br32d	1.00
F-17	UCLA	3	6.42	Br36d	1.00
F-18	UCLA	3	6.42	Br40d	1.00
F-19	UCLA	3	6.42	Br44d	1.00
F-20	UCLA	3	6.42	Br49d	1.00
F-21	UCLA	3	6.42	Br54d	1.00
F-22	UCLA	3	6.42	Br59d	1.00
F-23	UCLA	3	6.42	Br64d	1.00
F-24	UCLA	4	10.70	Br32d	1.00
F-25	UCLA	4	10.70	Br36d	1.00
F-26	UCLA	4	10.70	Br40d	1.00
F-27	UCLA	4	10.70	Br44d	1.00
F-28	UCLA	4	10.70	Br49d	1.00
F-29	UCLA	4	10.70	Br54d	1.00
F-30	UCLA	4	10.70	Br59d	1.00
F-31	UCLA	4	10.70	Br64d	1.00
F-32	UCLA	5	17.11	Br32d	1.00
F-33	UCLA	5	17.11	Br36d	1.00
F-34	UCLA	5	17.11	Br40d	1.00
F-35	UCLA	5	17.11	Br44d	1.00
F-36	UCLA	5	17.11	Br49d	1.00
F-37	UCLA	5	17.11	Br54d	1.00
F-38	UCLA	5	17.11	Br59d	1.00
F-39	UCLA	5	17.11	Br64d	1.00